

Math for Machine Learning and Deep Learning

Complete roadmap: 32 printable workbooks, hand-calculation style

How each workbook works (3 to 5 pages, A4)

- Page 1: short theory recap in plain language, with the key formulas.
- Next: one or two fully worked examples using small numbers.
- Then: blank-grid exercises to fill in with a pencil, from easy to harder.
- Last page: answer key. Tick its box in the list below when you finish a workbook.

Suggested pace: 2 to 3 workbooks a week, about 11 to 16 weeks in total. Modules build on each other, so follow the order.

Module 0: Warm-up (rebuild the basics)

☐	1	Exponents, logarithms and e	Powers and roots, log rules, why e shows up in sigmoid, softmax and cross-entropy
☐	2	Functions, graphs, slopes and Sigma notation	Read a graph, find a slope, expand and evaluate sums
☐	3	Vectors and matrices from scratch	Read shapes (rows x columns), add, scale, index entries

Module 1: Linear algebra

☐	4	Vectors, dot product, norms	Dot products, lengths, cosine similarity on small vectors
☐	5	Matrix multiplication	3x4 by 4x2 products on a grid; the core operation of deep learning
☐	6	Transpose, inverse, determinant	2x2 and 3x3 by hand
☐	7	Linear transformations and change of basis	See what a matrix does to points
☐	8	Eigenvalues and eigenvectors	Solve 2x2 cases by hand
☐	9	SVD, low-rank approximation and PCA	Compress a small matrix; find the main direction of a point cloud

Module 2: Calculus

☐	10	Derivatives and partial derivatives	Slope as rate of change; differentiate common functions
☐	11	Gradients and the chain rule	Chain rule on nested functions; gradient of a loss
☐	12	Jacobian and Hessian	Matrices of derivatives for small functions

Module 3: Optimization

☐	13	Gradient descent and learning rate	Take 5 steps by hand; see too-large vs too-small steps
☐	14	SGD, mini-batches and momentum	Compare update paths on a tiny dataset
☐	15	RMSProp and Adam	One full Adam update, every number filled in

Module 4: Probability and statistics

[]	16	Probability, conditional probability, Bayes	Tables and trees; Bayes' rule on a medical-test style problem
[]	17	Expectation, variance, key distributions	Bernoulli, Gaussian; compute mean and variance
[]	18	Maximum likelihood estimation	Fit a coin and a Gaussian by hand
[]	19	Entropy, cross-entropy, KL divergence	Compute all three for small distributions

Module 5: Classical machine learning

[]	20	Linear regression	Normal equation and one gradient descent step
[]	21	Logistic regression, sigmoid, softmax	Turn scores into probabilities; compute the loss

Module 6: Neural networks

[]	22	Forward pass and activation functions	ReLU, sigmoid, tanh through a 2-layer network
[]	23	Loss functions	MSE and cross-entropy, side by side
[]	24	Backpropagation by hand	Every gradient in a tiny network, layer by layer
[]	25	BatchNorm, LayerNorm and dropout	Normalize a small batch; apply a dropout mask

Module 7: Architectures

[]	26	Convolution and pooling	Slide a 3x3 filter over a 5x5 image; max pooling
[]	27	RNN and LSTM	Unroll 3 time steps; LSTM gates
[]	28	Embeddings	Lookup tables and similarity
[]	29	Attention and self-attention	Q, K, V and softmax on a 3-token sequence
[]	30	Transformer block and positional encoding	Assemble the pieces into one block

Module 8: Modern topics

[]	31	LoRA and quantization	Low-rank update of a weight matrix; round weights to fewer bits
[]	32	Diffusion basics	Add noise step by step; the idea of learning to remove it

Workbooks are original exercises in the spirit of hand-computing small numbers on grids. Topics can be reordered, trimmed or added to at any time.